A statement has a definite truth value: true or false.

Symbol	Read as	True when
¬ p	not p	p is false
p∧ q	p and q	both are true
p∨ q	p or q	at least one is true
p⊕ q	p exclusive-or q	exactly one is true
p→ q	if p then q	p is false, or q is true
p↔ q	p if and only if q	p and q match
∀ x P(x)	for every x, P(x)	P holds for all values in the domain
∃ x P(x)	there exists an x such that P(x)	P holds for at least one value

p → q is false only if p is true and q is false.

Form	Symbolic	Relation to p→ q
Contrapositive	¬ q→¬ p	equivalent
Converse	q→ p	not equivalent
Inverse	¬ p→¬ q	not equivalent

Valid means the conclusion is true whenever all premises are true.

Name	Form
Modus ponens	p→ q, p, ∴ q
Modus tollens	p→ q, ¬ q, ∴¬ p
Generalization	p, ∴ p∨ q
Specialization	p∧ q, ∴ p
Elimination	p∨ q, ¬ q, ∴ p
Transitivity	p→ q, q→ r, ∴ p→ r
Division into cases	p∨ q, p→ r, q→ r, ∴ r
Conjunction	p, q, ∴ p∧ q
Contradiction	¬ p→ℂ, ∴ p

Negate from outside: flip each quantifier, then negate the predicate.

Form	Statement	Equivalent?
Contrapositive	∀ x (¬ Q(x)→¬ P(x))	yes
Converse	∀ x (Q(x)→ P(x))	no
Inverse	∀ x (¬ P(x)→¬ Q(x))	no

∀x∃y lets y depend on x; ∃y∀x needs one y for every x.

Equivalent forms have the same truth value in every truth-table row.

Name	Equivalence
De Morgan	¬(p ∧ q) ≡ ¬p ∨ ¬q; ¬(p ∨ q) ≡ ¬p ∧ ¬q
Commutative	p∧ q≡ q∧ p, and the same for ∨
Associative	(p∧ q)∧ r≡ p∧(q∧ r), and the same for ∨
Distributive	p∧(q∨ r)≡(p∧ q)∨(p∧ r), and p∨(q∧ r)≡(p∨ q)∧(p∨ r)
Identity	p∧𝕥≡ p, and p∨ℂ≡ p
Negation	p∨¬ p≡𝕥, and p∧¬ p≡ℂ
Double negation	¬(¬ p)≡ p
Idempotent	p∧ p≡ p, and p∨ p≡ p
Universal bound	p∨𝕥≡𝕥, and p∧ℂ≡ℂ
Absorption	p∨(p∧ q)≡ p, and p∧(p∨ q)≡ p
Negations of 𝕥 and ℂ	¬𝕥≡ℂ, and ¬ℂ≡𝕥

The converse and inverse are equivalent to one another.

English	Means
p only if q	p→ q
q is necessary for p	p→ q
p is sufficient for q	p→ q
p if and only if q	(p→ q)∧(q→ p)
p unless q	¬ q→ p

A predicate becomes a statement when x has a value or quantifier.

English	Form
All A are B	∀ x (A(x)→ B(x))
Some A are B	∃ x (A(x)∧ B(x))
No A are B	∀ x (A(x)→¬ B(x))
Some A are not B	∃ x (A(x)∧¬ B(x))

A sufficient for B and B necessary for A mean ∀x(A(x) → B(x)).

Quantifier negation
¬∀x P(x) ≡ ∃x ¬P(x)
¬∃x P(x) ≡ ∀x ¬P(x)
¬∀x (P(x) → Q(x)) ≡ ∃x (P(x) ∧ ¬Q(x))

Two quantifiers
∀x ∃y P(x,y): y may depend on x
∃y ∀x P(x,y): one y works for every x
∃y ∀x P(x,y) implies ∀x ∃y P(x,y), but not conversely
¬(∀x ∃y P(x,y)) ≡ ∃x ∀y ¬P(x,y)